

Problem

Submissions

Leaderboard

Discussions

Euclid's Identity states that for any two positive integers A and B, there exist integers x and y such that: $Ax + By = D$

where D is the Greatest Common Divisor (GCD) of A and B.

Given two positive integers A and B, find: $x \ y \ D = \text{gcd}(A, B)$

such that:

$Ax + By = D$

If multiple pairs (x, y) satisfy the equation, choose the pair with the minimum value of: $|x| + |y|$

If there is still more than one pair, choose the pair for which:

$x \leq y$

Input Format

A B

Constraints

$1 \leq A, B \leq 10^9$

Output Format

$x \ y \ D$ D is $\text{gcd}(A, B)$ $Ax + By = D$ (x, y) satisfies the minimum-selection rules.

Sample Input 0

4 6

Sample Output 0

-1 1 2

Change Theme

Language

Java 15

↺

⋮

```
1
2 import java.util.*;
3 import java.io.*;
4 public class Solution{
5     public static void main(String args[]){
6         Scanner sc=new Scanner(System.in);
7         int a=sc.nextInt();
8         int b=sc.nextInt();
9         int s1=1,s2=0;
10        int t1=0,t2=1;
11        int s=0,t=0;
12        while(b!=0){
13            int q=a/b;
14            int r=a%b;
15            s=s1-(s2*q);
16            t=t1-(t2*q);
17            a=b;
18            b=r;
19            s1=s2;
20            s2=s;
21            t1=t2;
22            t2=t;
23        }
24        System.out.println(s1 + " " + t1 + " ")
25    }
```

Line: 27 Col: 1

Upload Code as File

Run Code

Submit Code

☐ Test against custom input

✔ Test case 0

Compiler Message

Success